Rockot / Breeze-KM from Plesetsk

Ascent trajectory model of the User's Guide reference mission:
650 km circular orbit, 86.58° inclination | AeroSim | October 2026

The model flies the guide's reference mission to orbit

1,570 kg

payload to 650 km circular

User's Guide fig. 3-1, i = 86.4°

64.8 kPa

max dynamic pressure at 51 s

guide: 61.29 kPa at 50 s

1,038 s

lift-off to transfer orbit

guide: 882 s (end of Breeze-KM burn 1)

- Stage 1 and 2 burnout speeds within -0.2 % and +0.5 % of the guide's trajectory (fig. 3-6).
- Breeze-KM burn 1 reaches a 160 x 635 km transfer orbit (guide: 160 x 635 km); burn 2 circularises at 643 x 666 km, i = 86.34°.
- Stage 1/2 propellant masses are not published; they were fitted to the guide's waypoints. The fitted vehicle weighs 117.0 t at lift-off against the guide's 107 t.

Three hypergolic stages; the top one restarts

Stage 1

3 x RD-0233 + RD-0234

Thrust, vacuum	**2,178 kN**
Isp, vacuum	**310.0 s**
Burn time	**121 s**
Propellant	**86,691 kg**
Dry mass	**5,695 kg**

converted SS-19 stage, 17.2 m long

Stage 2

RD-0235 + 4 RD-0236 verniers

Thrust, vacuum	**248 kN**
Isp, vacuum	**320.0 s**
Burn time	**183 s**
Propellant	**14,475 kg**
Dry mass	**1,485 kg**

3.9 m long, verniers steer

Breeze-KM

S5.98M, 20 kN

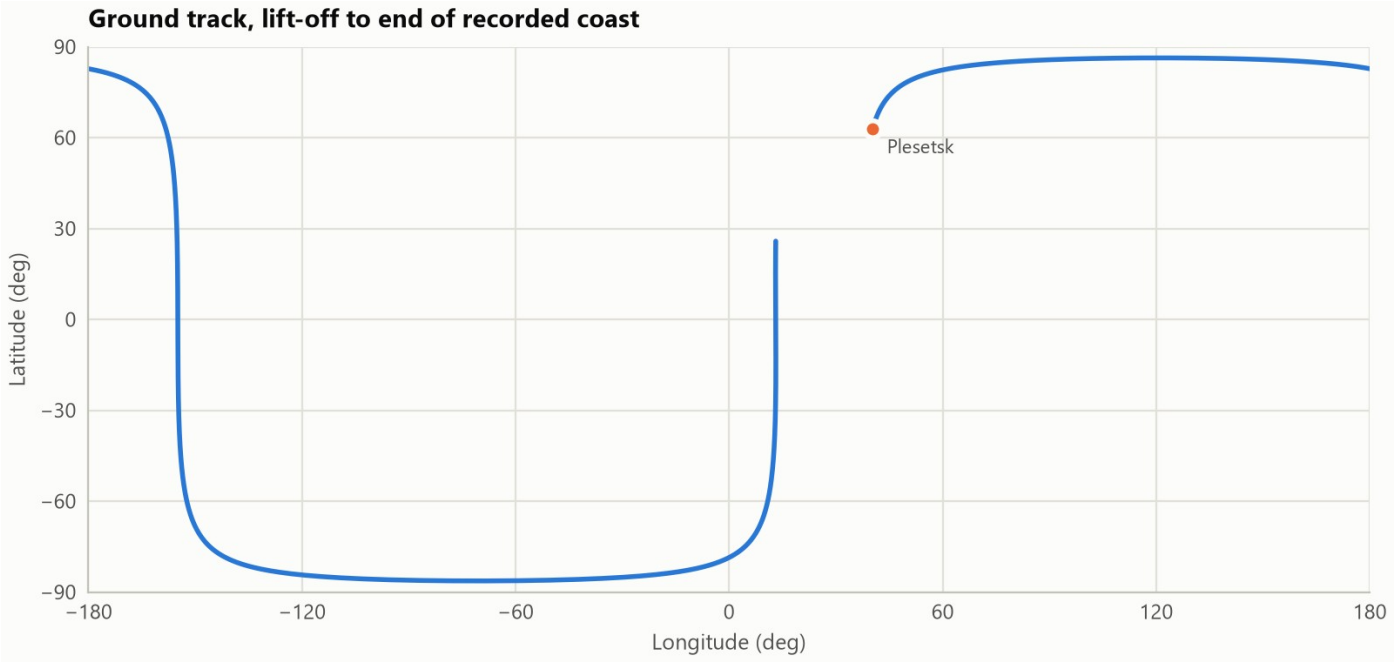
Thrust, vacuum	**20 kN**
Isp, vacuum	**325.5 s**
Burn time	**up to 1,000 s**
Propellant	**4,975 kg**
Dry mass	**1,320 kg**

up to 8 ignitions, 12 x 13 N + 4 x 400 N thrusters

Isp, burn times and Breeze-KM data: User's Guide tables 2-2 to 2-8. Stage 1/2 propellant is fitted (slide 9), so their thrust follows from it at the guide's burn time; the guide quotes 2,070 kN and 256 kN vacuum. Dry masses: Encyclopedia Astronautica.

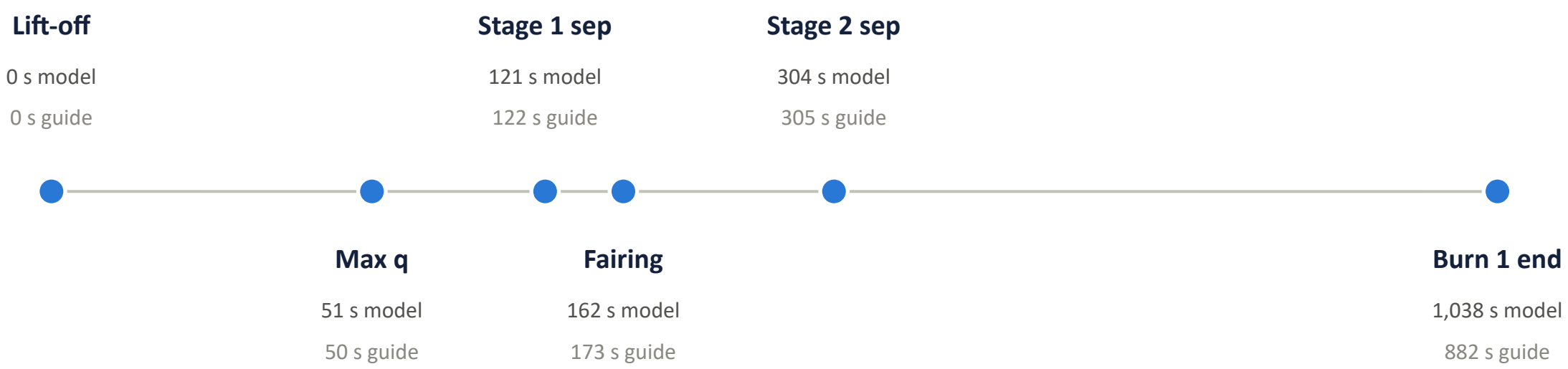
Launch from Plesetsk, northbound into a near-polar orbit

- Plesetsk Cosmodrome, 62.8° N 40.3° E, about 200 km south of Archangel (guide §3.2).
- Allowed azimuths from 4.8° (i = 86.4°) to 76.2° (i = 63.2°); 341.5° for sun-synchronous orbits.
- Target plane i = 86.583°: the model launches along the rotation-corrected azimuth and yaws out residual out-of-plane velocity.
- Ground track runs north over the Barents Sea and the pole, then down the far side of the Earth.



AeroSim model output (aerosim.plots.ground_track)

About fifteen minutes from lift-off to the transfer orbit

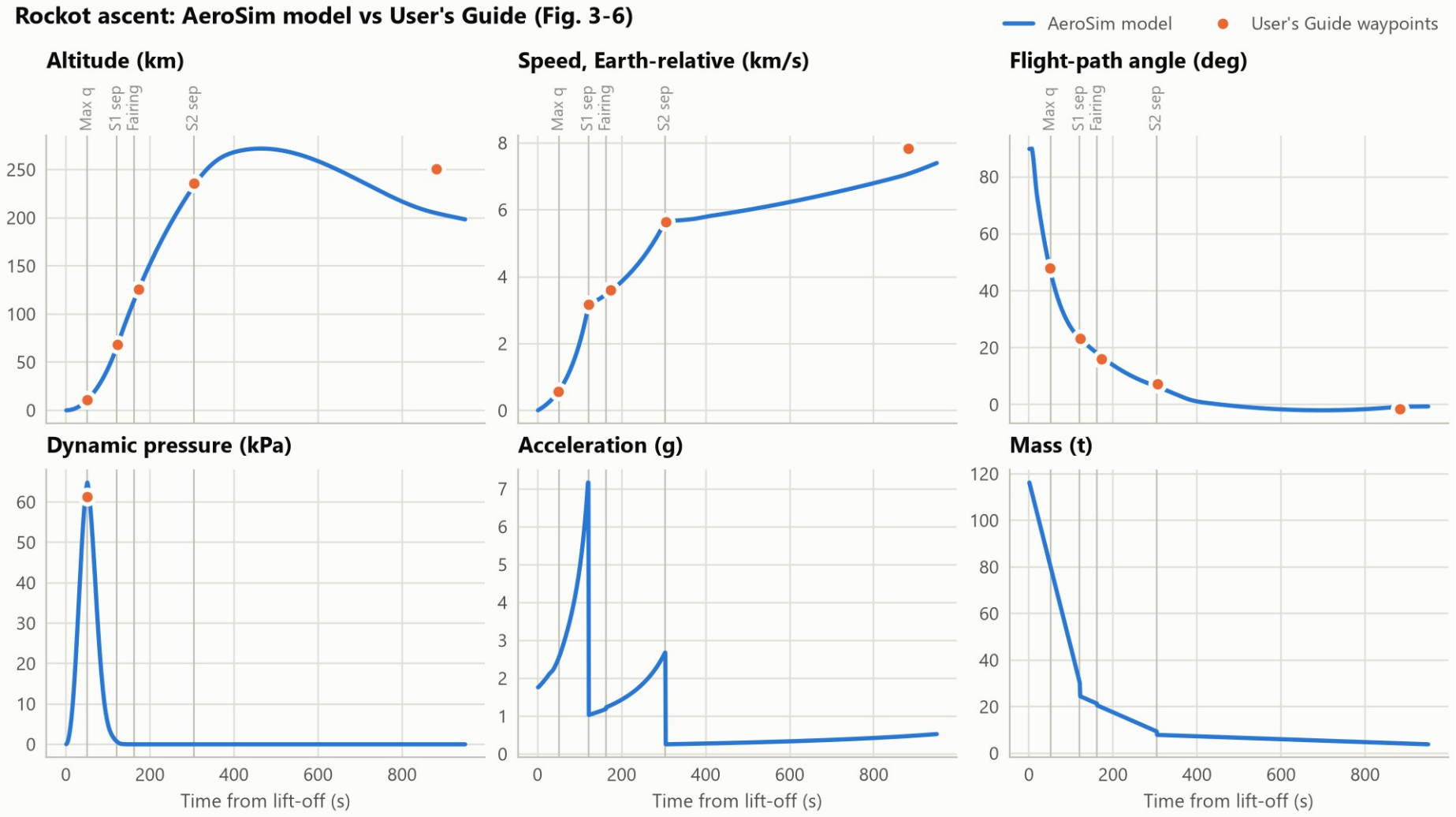


Time axis is square-root scaled so the first two minutes are readable.

Then: coast to apogee, Breeze-KM burn 2 at 4,093 s for 22 s (guide: 4,360 s, about 39 s) circularises the orbit; the guide deploys the spacecraft at 5,861 s.

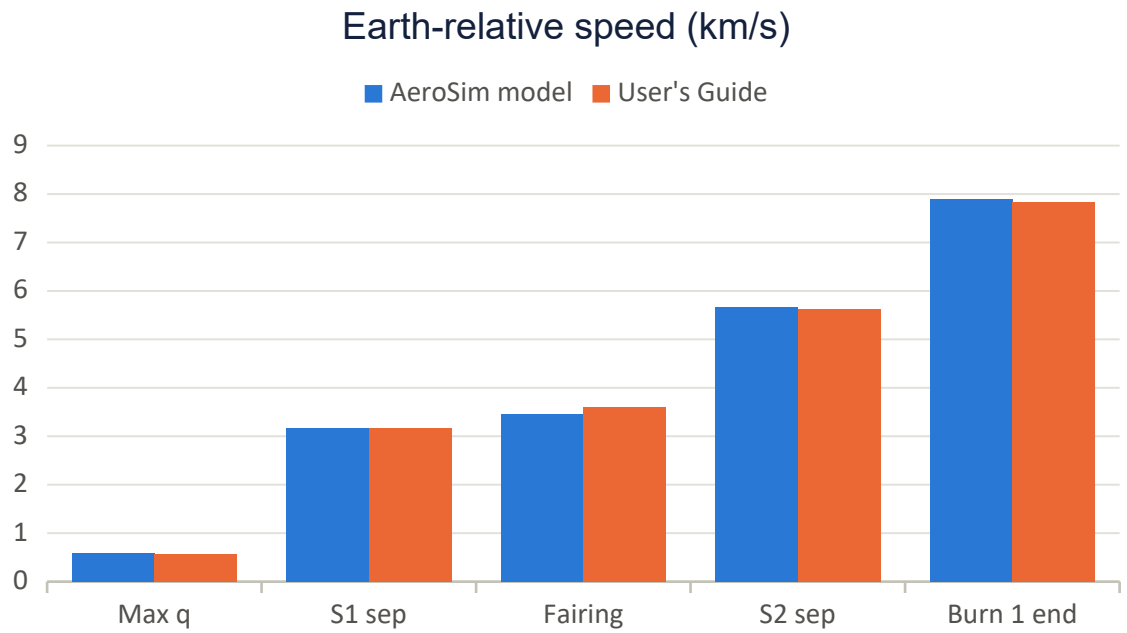
Rockot / Breeze-KM ascent model | AeroSim | data: Rockot User's Guide EHB0003 Iss. 5 (2011)

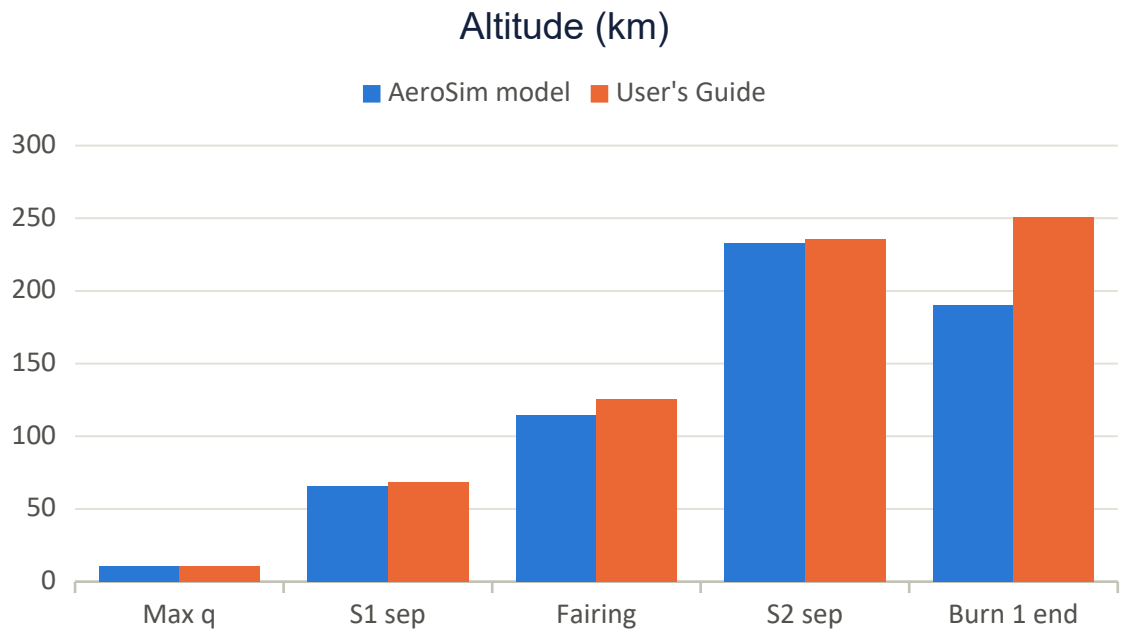
Ascent profile tracks the guide's published waypoints



- Blue line: AeroSim model. Orange points: User's Guide fig. 3-6 waypoints.
- Max q, both stagings and the end of burn 1 sit on or near the guide's points.
- Peak axial acceleration 7.2 g at stage 1 burnout; guide table 5-1 gives 6.2-8.0 g.
- Stage 2 peaks at 2.7 g (guide: up to 3.0 g).

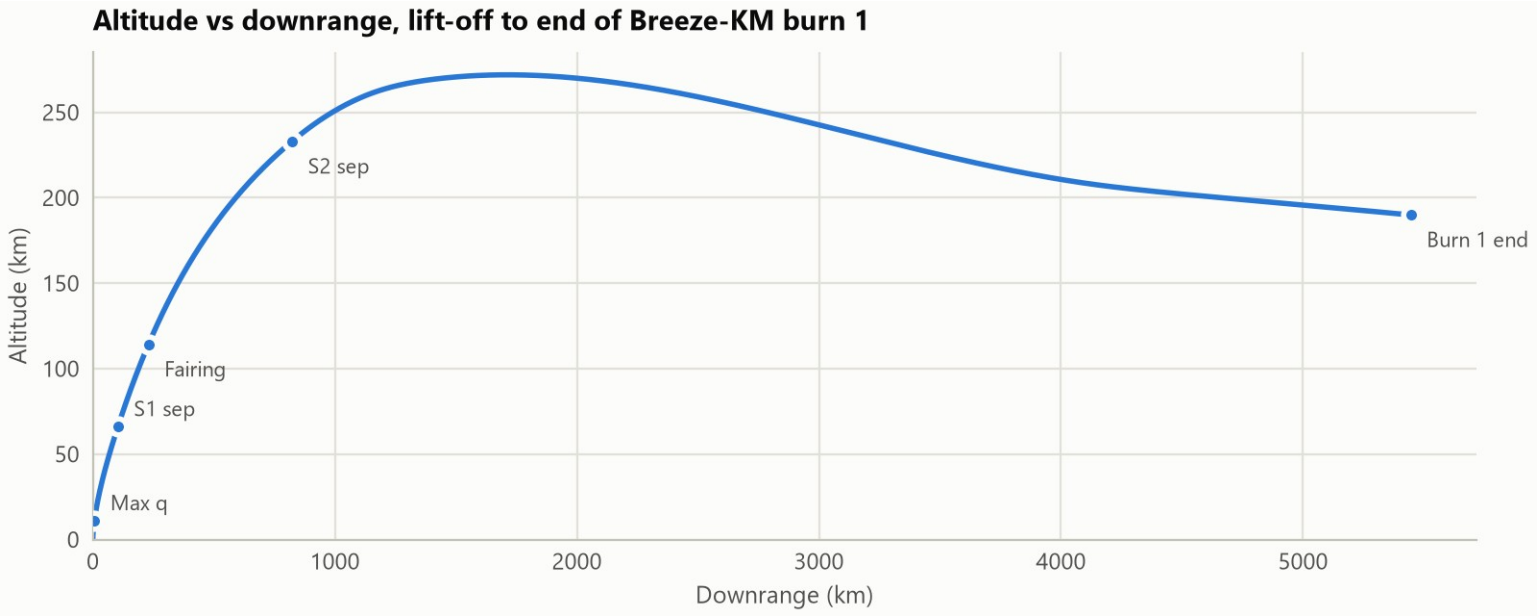
Model against the guide at each event





Event	Time model / guide (s)	Altitude (km)	Speed (m/s)	Flight-path angle (deg)
Max q	51 / 50	10.9 / 10.7	591 / 567	46.9 / 47.9
S1 sep	121 / 122	65.9 / 68.3	3,156 / 3,163	23.3 / 23.3
Fairing	162 / 173	114.2 / 125.6	3,464 / 3,599	18.0 / 16.0
S2 sep	304 / 305	233.0 / 235.7	5,663 / 5,634	6.5 / 7.3
Burn 1 end	1,038 / 882	189.9 / 250.7	7,902 / 7,832	-0.6 / -1.5

Breeze-KM burns twice to reach the 650 km class orbit



	Perigee	Apogee	Inclination
Transfer orbit, model	160 km	635 km	86.34°
Transfer orbit, guide	160 km	635 km	86.58°
Final orbit, model	643 km	666 km	86.34°
Deployment orbit, guide	650 km	650 km	86.58°

Insertion

- Burn 1 is closed-loop: pitch holds a commanded climb rate while the 20 kN engine builds speed.
- Burn 2 fires at apogee, prograde, and stops when perigee reaches the transfer apogee.
- Burn 1 runs 734 s against the guide's 577 s and ends lower (190 km vs 251 km): the model's steering is less efficient.
- Breeze-KM propellant left: 235 kg of 4,975 kg.

Two masses and five steering settings were fitted

Quantity	Source value	Used in model	Basis
Stage 1 propellant	71,455 kg (Astronautix)	86,691 kg	fitted
Stage 2 propellant	10,710 kg (Astronautix)	14,475 kg	fitted
Breeze-KM propellant / dry	4,975 / 1,320 kg	4,975 / 1,320 kg	User's Guide t. 2-8
Payload	1,570 kg	1,570 kg	User's Guide fig. 3-1
Fairing	not published	700 kg	estimate
Lift-off mass	107 t	117.0 t	result

- The guide gives engines, burn times, Breeze-KM masses and lift-off mass, but not stage 1/2 propellant.
- Published stage masses leave stage 1 about 300 m/s slow at burnout against the guide's own trajectory.
- A least-squares fit adjusts both propellant loads plus pitch kick, stage 2 pitch and Breeze-KM climb rate to ten guide targets.
- Mass flow is propellant / burn time at constant thrust, so thrust scales with the fitted load.

8 of 10 targets met

within 1.5 tolerance units (25 m/s speed, 2-4 km altitude, 1° angle, 3 km perigee)

Missed: Breeze-KM burn 1 ends 156 s later than the guide, and the fitted vehicle is 10.0 t heavier at lift-off. Both point the same way: the model needs more energy than the real vehicle to fly the same trajectory, most likely from its simple insertion steering and generic drag.

Limits and next steps

What the model leaves out

- 3-DOF point mass: no attitude dynamics or control-loop limits
- Generic slender-body drag curve, not Rockot wind-tunnel data
- Constant thrust per stage; no throttling, tail-off or vernier phases
- Spherical-harmonic gravity to J2 only; exponential upper atmosphere

Next steps

- Add stage drop zones and impact points (guide: 990-1,100 km for stage 1)
- Payload-vs-orbit sweep to reproduce guide fig. 3-1
- SSO mission: 341.5° azimuth, parking orbit and plane change
- Monte Carlo on thrust, Isp and winds for dispersion

Sources and reproduction

- Eurockot Launch Services, Rockot User's Guide, EHB0003, Issue 5, Revision 0, August 2011: tables 2-1 to 2-8, section 3, figures 3-1 and 3-6. Copy in docs/references/rockot/.
- Encyclopedia Astronautica, Rokot (astronautix.com/r/rokot.html): stage gross and empty masses.
- D. Vallado, Fundamentals of Astrodynamics and Applications, table 8-4: upper-atmosphere density.
- Model: AeroSim configs/examples/rocket/orbital/rockot.yaml; calibration: examples/rockot_ascent.py.
- This briefing: python docs/briefings/rockot/make_briefing.py, then node docs/briefings/rockot/build/deck.js.